

Scalability & Future Plan

Date	3 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users	System performance may degrade with many students accessing the platform simultaneously	High
2	No integration with school or college Learning Management Systems (LMS)	Restricts adoption within existing classroom and institutional workflows	High
3	No long-term storage of emotion trends across semesters	Students and teachers cannot track long-term emotional learning patterns	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a single local Streamlit server	Deploy on cloud infrastructure (AWS, Google Cloud, Streamlit Community Cloud) with load balancing and auto-scaling
2	Data Storage	Basic CSV file storage for session history and analytics	Implement scalable databases such as MongoDB or PostgreSQL with cloud storage support
3	Performance	BERT and BiLSTM inference run in real time for each request	Use model caching, batching, and optimized inference pipelines (e.g., quantized models)
4	Security	Basic local storage with no user authentication	Implement OAuth-based login, encrypted storage, and role-based access control (Student / Teacher)

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multilingual emotion detection and voice-based emotion recognition	2 Months	Expands accessibility to non-English speakers and enables hands-free interaction
Phase 3	Facial emotion recognition and Learning Management System (LMS) integration	4 Months	Improves emotion detection accuracy and enables adoption within existing classroom platforms
Phase 4	Teacher dashboard, adaptive learning paths, and mobile application	6 Months	Expands accessibility, supports personalized learning at scale, and enables broader deployment